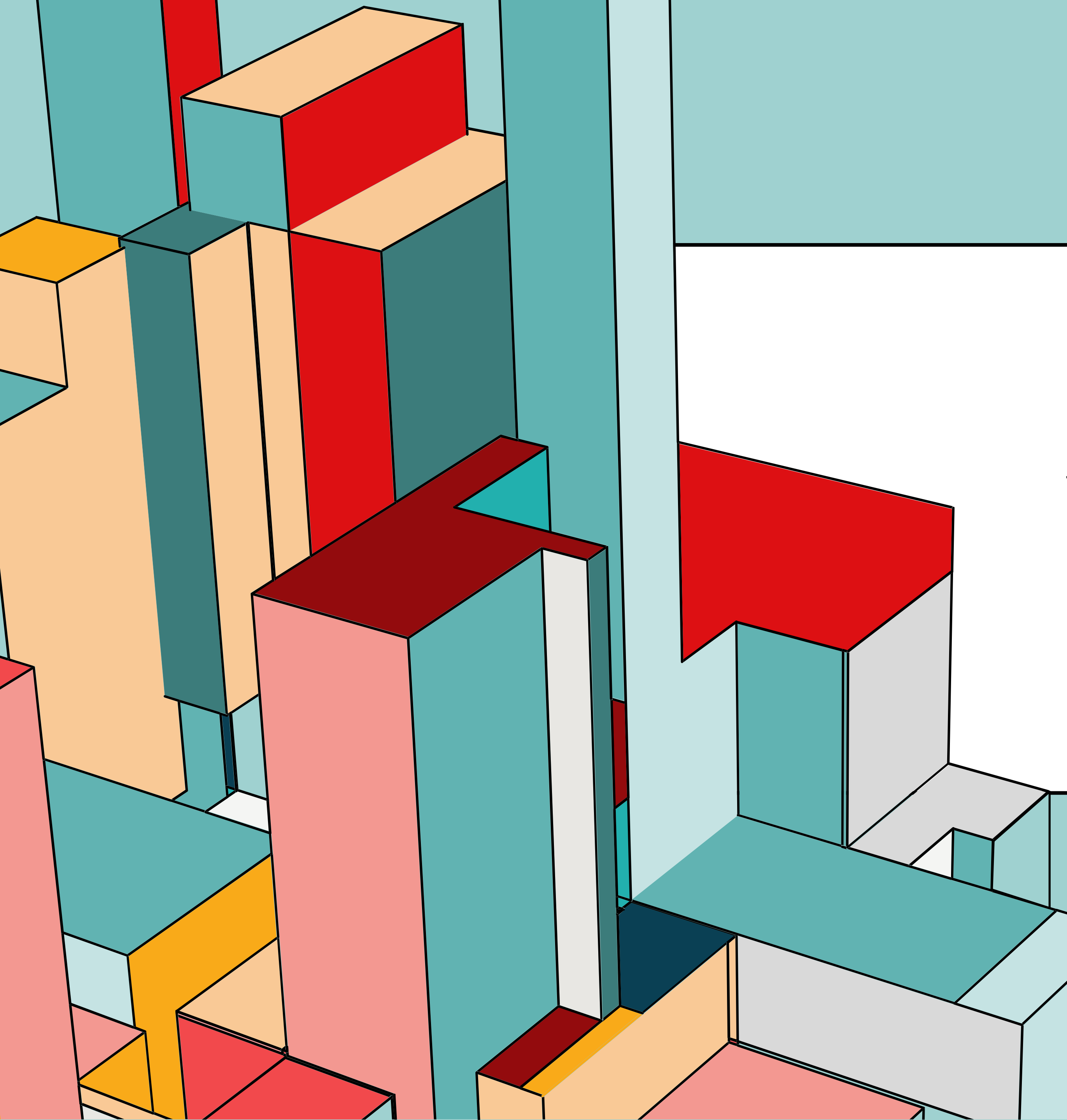


# RANDOM VARIABLE

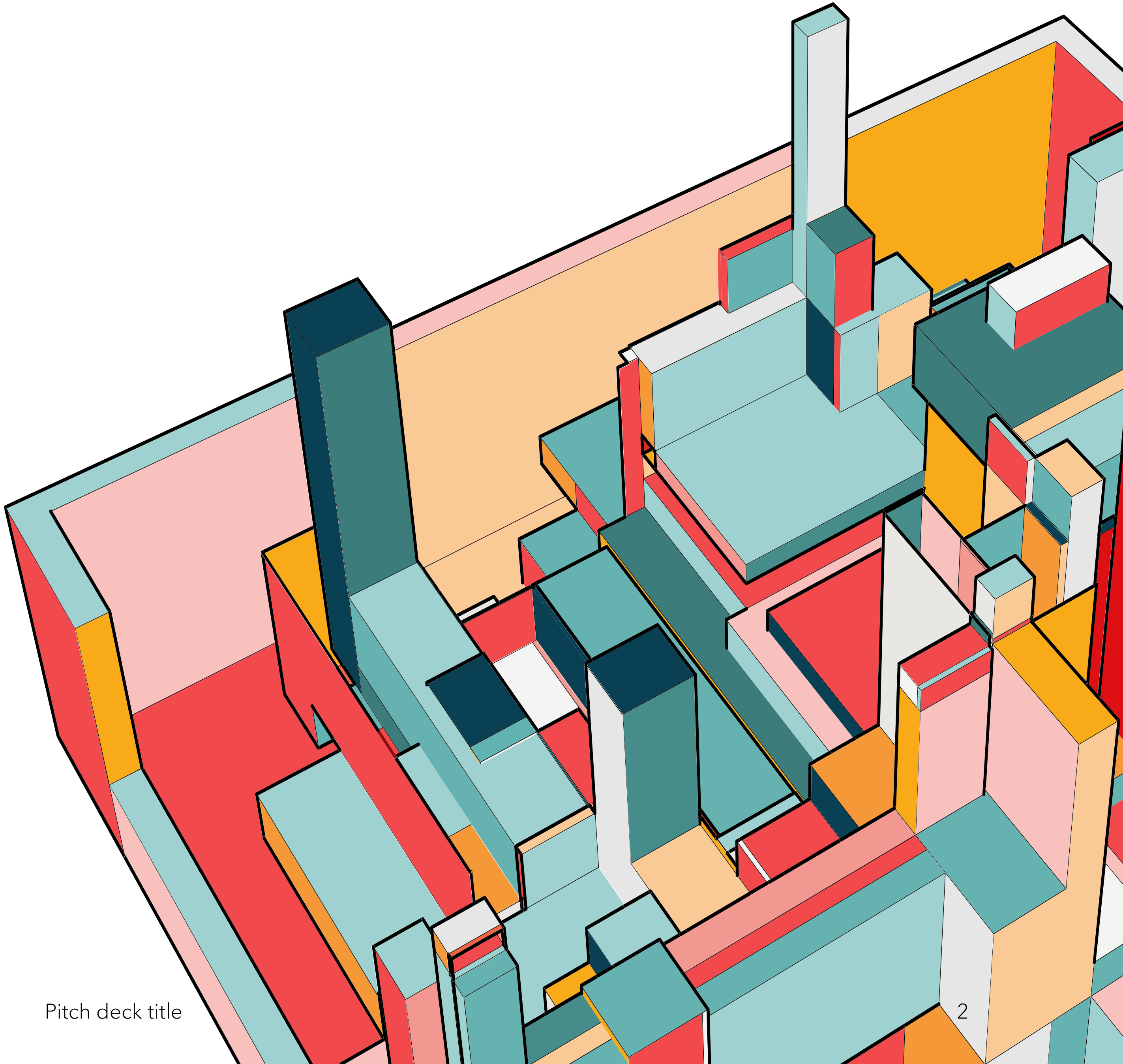
STATISTICS AND PROBABILITY

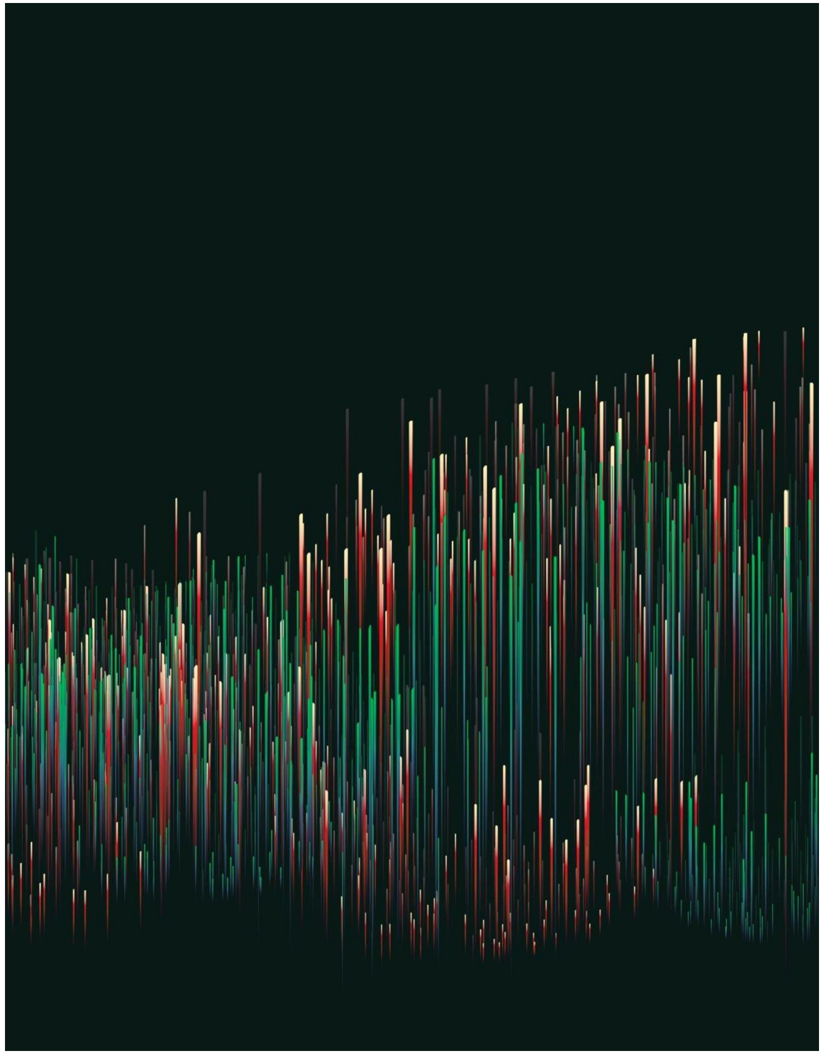
3RD Quarter

# LEARNING OBJECTIVES

❖ Illustrates a random variable (discrete and continuous).

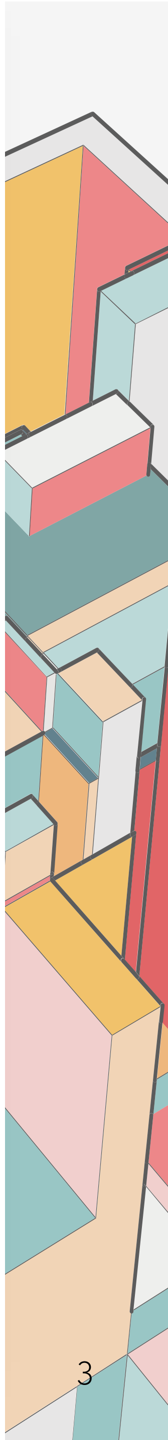
❖ Distinguishes between a discrete and a continuous random variable.

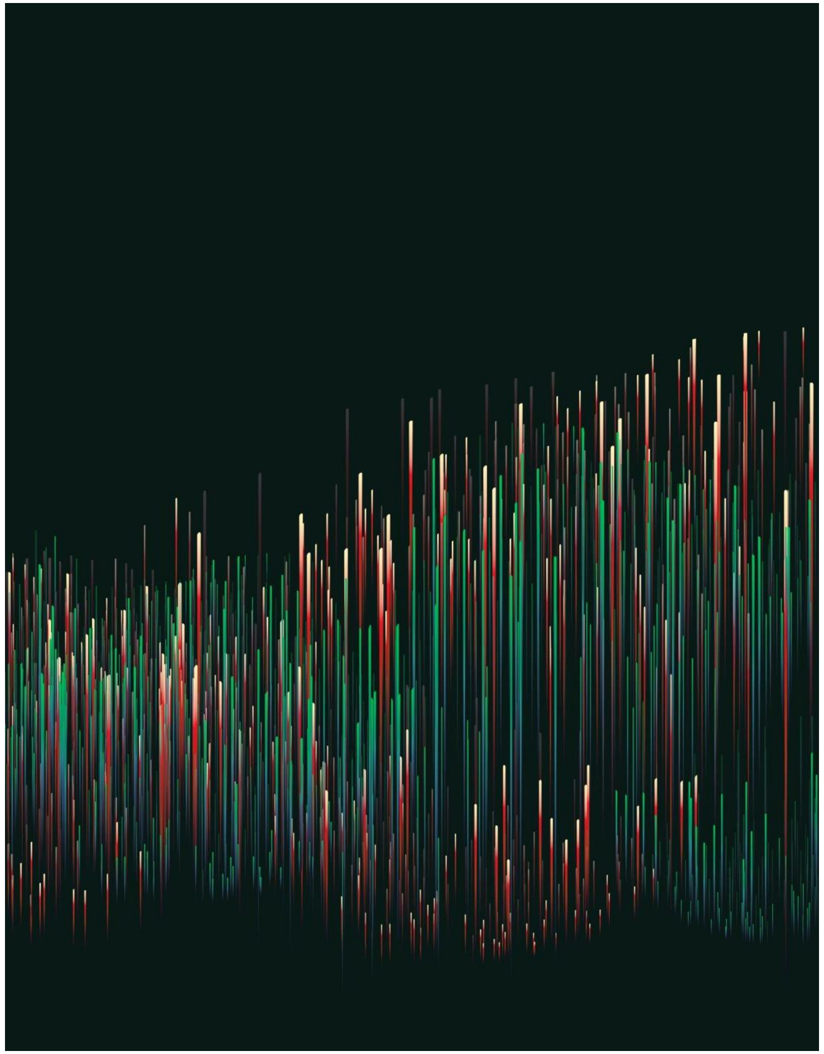




# GUIDE QUESTIONS:

- Do these concepts make sense in the real world?
- How can we apply these concepts to our day-to-day living?





# GUIDE QUESTIONS:

- Why do we need to study about random variables?

# SAMPLE SCENARIO 1:

- Imagine you are an avid online shopper, and you often order products from different online retailers. The ***time*** it takes for your ordered items to be delivered can be considered as a ***random variable***.

# SAMPLE SCENARIO 2:

- Every time it rains, PAGASA monitors rainfall by measuring the amount of precipitation to provide accurate weather updates and bulletins. The ***amount of rainfall*** is another example of a ***random variable***.

# SAMPLE SCENARIO 3:

- A student takes a 48-item multiple-choice test. The *number of correct answers* represent a kind of a *random variable*.

# PROCESS QUESTIONS:

How will you characterize a random variable based on the given situations?

# Random Variable

## DEFINITION

A **random variable** is a function that assigns a numerical value to each possible outcome of a random experiment. It bridges the outcomes of the experiment and numerical analysis.

# Random Variable

**DEFINITION**

A **random variable** $X$ is a function $X{:}S \rightarrow \Re$, where $S$ is the sample space of the random experiment (the set of all possible outcomes), and $\Re$ is the set of real numbers.

# PROCESS QUESTIONS:

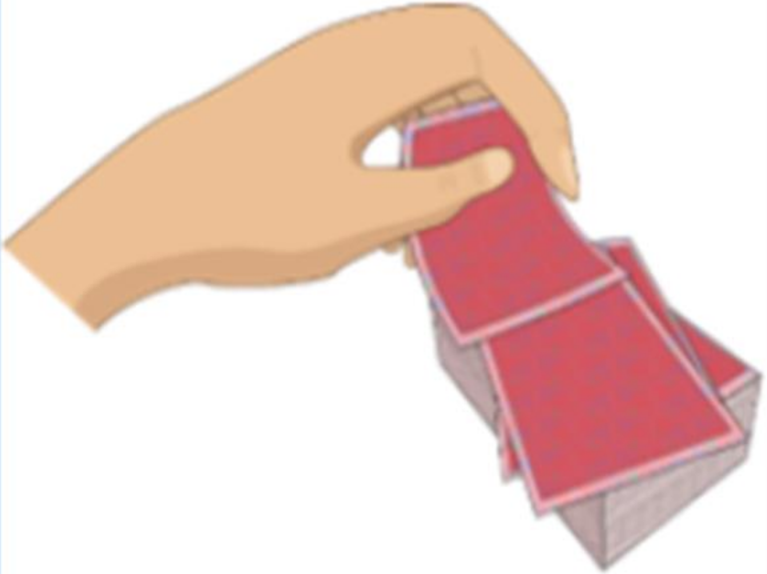
What are the other situations that show random variable?

# Random Experiments

Here are some examples of **random experiments.**

Tossing a coin

Rolling a die



Drawing a card from a deck

# Random Experiments

**DEFINITION**

- any process used to draw outcomes with the element of randomness.

Note: The actual outcomes are said to happen randomly, without specific rule or plan of certain determination, but have definite probability or chance of occurrence.

## Random Experiment

### DISCUSSION

- If a random experiment is to be performed, the possible outcomes can already be determined and enumerated but these outcomes are uncertain to happen.
- Although outcomes are uncertain, they can be determined by using probabilities. Commonly, outcomes of random experiments have equal probabilities of occurring.

# Random Experiments

Here are some examples of **random experiments.**

Tossing a coin

The outcomes can be heads or tails, but you can't predict which one will occur.

# Random Experiments

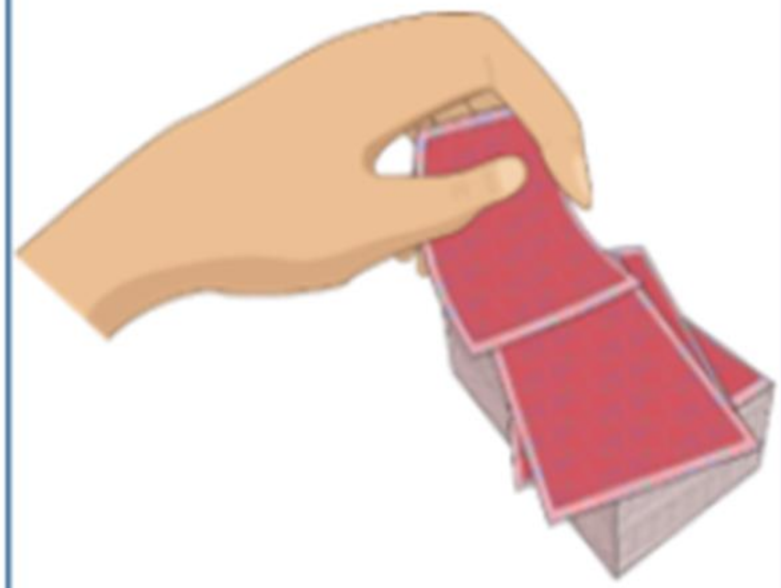
Here are some examples of **random experiments.**

Rolling a die

The outcomes can be any number from **1 to 6**, and each roll is unpredictable.

# Random Experiments

Here are some examples of **random experiments.**



**Drawing a card from a deck**

You know there are 52 cards, but you can't predict which card you'll get.

# IMPORTANT TERMS:

**Sample space:**
A sample space can be defined as the list of all possible outcomes of a random experiment.

**Outcome:**
An outcome is a possible result of the random experiment.

# IMPORTANT TERMS:

**Event:**

An event is a possible outcome of an experiment and forms a subset of a sample space.

**Trial:**

When a random experiment is repeated many times each one is known as a trial.

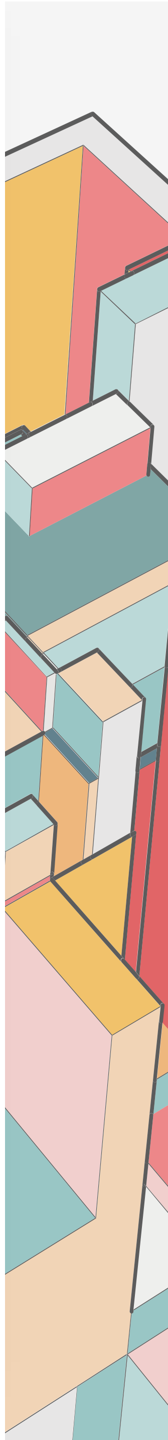
# EXAMPLE:

***Random Experiment:***

- Tossing 2 coins simultaneously

***Sample Space:***

- {head-head, head-tail, tail-head, tail-tail}



# EXAMPLE:

*Outcome*:

- Either a ***head-tail or any of the possible outcomes***

*Event:*

- Event *A*: Getting exactly one head

$$A = \{ HT, TH \}$$

- Event *B*: Getting exactly at least one tail

$$B = \{ HT, TH, TT \}$$

# GUIDE QUESTION:

What are the types of random variables and how do we classify them?

# TYPES OF RANDOM VARIABLES:

- DISCRETE Random Variable
- CONTINUOUS Random Variable

# DISCRETE RANDOM VARIABLE

A random variable is *discrete* if it has a *finite* or *countable* number of possible outcomes that can be listed.

# DISCRETE RANDOM VARIABLE:

Examples:

- The result of rolling a die
- The number of heads in tossing a coin

# DISCRETE RANDOM VARIABLE:

Characteristics:

- Values are typically integers
- Probability is assigned to each specific value, represented by a *probability mass function (PMF)*.

# CONTINUOUS RANDOM VARIABLE:

A random variable is called *continuous* if it can take on any value within a given range or interval, including an ***infinite*** number of possible values.

# CONTINUOUS RANDOM VARIABLE:

Examples:

- The height of students in class
- The time it takes to complete a task

# CONTINUOUS RANDOM VARIABLE:

Characteristics:

- Values are real numbers
- Probability is described over intervals rather than specific points, represented by a *probability density function (PDF)*.

# EXAMPLE

1

**Random experiment:**

Tossing a one-peso coin once

## Random Variable: X

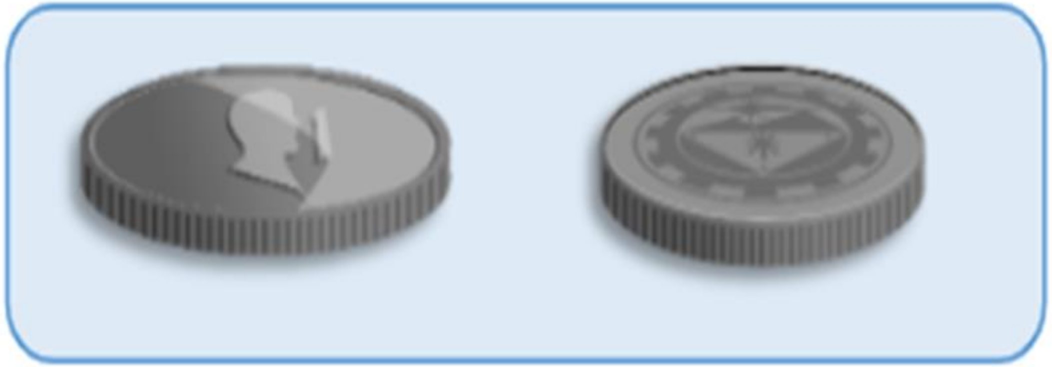
Let $X$ = occurrence of head

# EXAMPLE

1

When tossing a coin, there are only **two** possible outcomes; either the head or the tail.

**Possible outcomes**



**Head** **Tail**

# EXAMPLE

1

Let $X$ = occurrence of head

If tail turns up, it means there is **0 head.**

If head turns up there is **1 head.**

**Possible values of X** 0, 1

**Therefore,** $X = \{0, 1\}$

DISCRETE

# EXAMPLE

**Random experiment:**

Rolling a die once

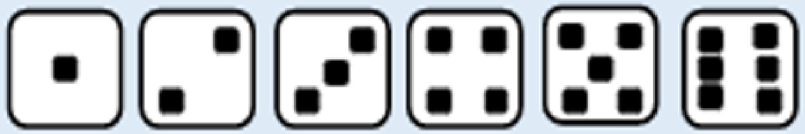
**Random Variable: Y**

Let $Y$ = the number of dots on the top face of the die

# EXAMPLE

Let $Y$ $=$ the number of dots on the top face of the die

When we roll a die, there are six distinct possible outcomes:



# EXAMPLE

If we roll a die, there are different number of dots that are shown in each face.

**Possible values of Y**

1, 2, 3, 4, 5, and 6

DISCRETE

**Therefore,**

$$Y = \{1, 2, 3, 4, 5, 6\}$$

2

# EXAMPLE

3

**Random experiment:**

Mrs. Alberca was given a task to get the weight of some Grade 11 students.

Random Variable: ***A***

**Let A = weight of every Grade 11 student**

# EXAMPLE

3

A = “Weight of every Grade 11 student”

x = student number

| x | A(x) = Weight (in kg) |
|---|---|
| 1 | 43 |
| 2 | 53.12 |
| 3 | 46.77 |
| 4 | 56 |
| 5 | 60 |
| 6 | 50.08 |

# EXAMPLE

## CONTINUOUS

3

$$A = \{43, 53.12, 46.77, 56, 60, 50.08\}$$

The observed weights vary from person to person in the class. The lightest student in their class weighs 43 kg and the heaviest 60 kg.

$\boxed{MIN = 43}$ $\boxed{MAX = 60}$

# EXAMPLE

4

**Random Experiment:**

Measuring the time (in seconds) required by a randomly chosen runner to complete one lap.

B = "Time required to finish a lap"

| x | B(x) = Time (in min.) |
|---|---|
| 1 – Gina | 4.5 |
| 2 – Timmy | 5.25 |
| 3 – Polly | 6 |
| 4 – Rina | 4.67 |
| 5 – Martel | 3.87 |

# EXAMPLE

## CONTINUOUS

4

**B = { 4.5, 5.25, 6, 4.67, 3.87 }**

The time required to finish a lap vary from person to person. The fastest runner can finish a lap in 3.87 minutes and the slowest in 6 minutes.

# CHECK FOR UNDERSTANDING


Are there any questions or clarifications?

# SUMMARY

- **Random experiment** is any process used to draw outcomes with the element of randomness
- **Random variable** is a function that assigns numerical value/s to the possible outcomes that will result from a random experiment
- **Discrete random variable** takes on a set of distinct values that are listable. It has finite or infinite number of possible values.

# SUMMARY

- **Random experiment** is any process used to draw outcomes with the element of randomness
- **Random variable** is a function that assigns numerical value/s to the possible outcomes that will result from a random experiment
- **Discrete random variable** takes on a set of distinct values that are listable. It has finite or infinite number of possible values.

# SUMMARY

- **Continuous random variable** takes on any value within a specific interval and holds an uncountable infinite number of possible values.
- In general, you can determine if a variable is continuous if possible values of the variable can be obtained by measuring and not by counting.

# EXAMPLE

1

**Discrete random variable**

Al cannot decide if he should wear his favorite shirt today so he chose to depend his decision on a toss coin. He thought of playing it five times and if more heads appear than tails, he will wear it.

# SOLUTION *(continuation)*

1

X = number of times head will appear

$X = \{0, 1, 2, 3, 4, 5\}$

The number of heads could be any integer value between 0 and 5.

If $X \geq 3$, Al will wear his favorite shirt tonight.

# SOLUTION *(continuation)*

1

X is a **discrete random variable** because...

Although the values of X could be any integer value between 0 and 5, **they could not be just any number** within that interval so there cannot be 3.5 heads.

There can only be **distinct values** possible which are 0, 1, 2, 3, 4, and 5.

# Continuous Random Variable

**DEFINITION**

A **continuous random variable** can take on any value within a specific interval and has an uncountable or infinite number of possible values.

# EXAMPLE

2

**Continuous random variable**

The city sports meet this year mandates all runners to be 40 kg to 65 kg in weight. Otherwise, they will not qualify to join any running event.

All the names of qualified athletes were listed down. Their weights are then measured and recorded.

# SOLUTION *(continuation)*

2

| Runner # (x) | Weight in kg A(x) |
|---|---|
| 1 | 43 |
| 2 | 54.02 |
| 3 | 39.76 |
| 4 | 51 |
| 5 | 63.5 |

A(x) = weight of runner x in kg

The variable A can take any value within the range of 40 to 65.

# SOLUTION *(continuation)*

2

A is a **continuous random variable** because...

The variable A can take any value within the range of 40 to 65. Some runners may weigh exactly 40 or 65 or even other integer value within that range. However, some may weigh 54.02 or 54.5. There are infinitely many possible values of A that's within the set range.

# Discrete vs. Continuous

## EXAMPLES

**Discrete**

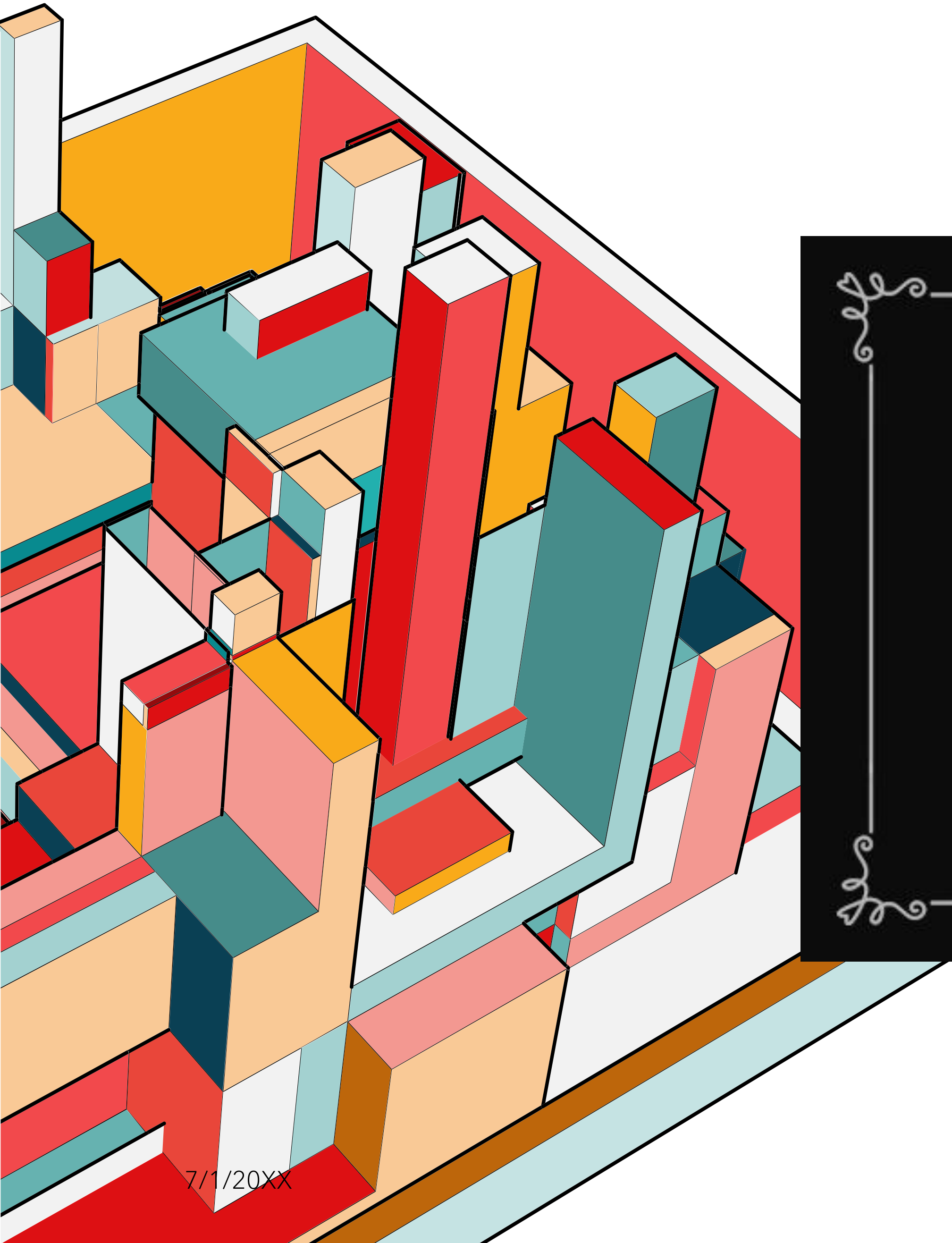
- number of heads in a toss coin
- number of patients with blood type O
- whether one's bet wins or not

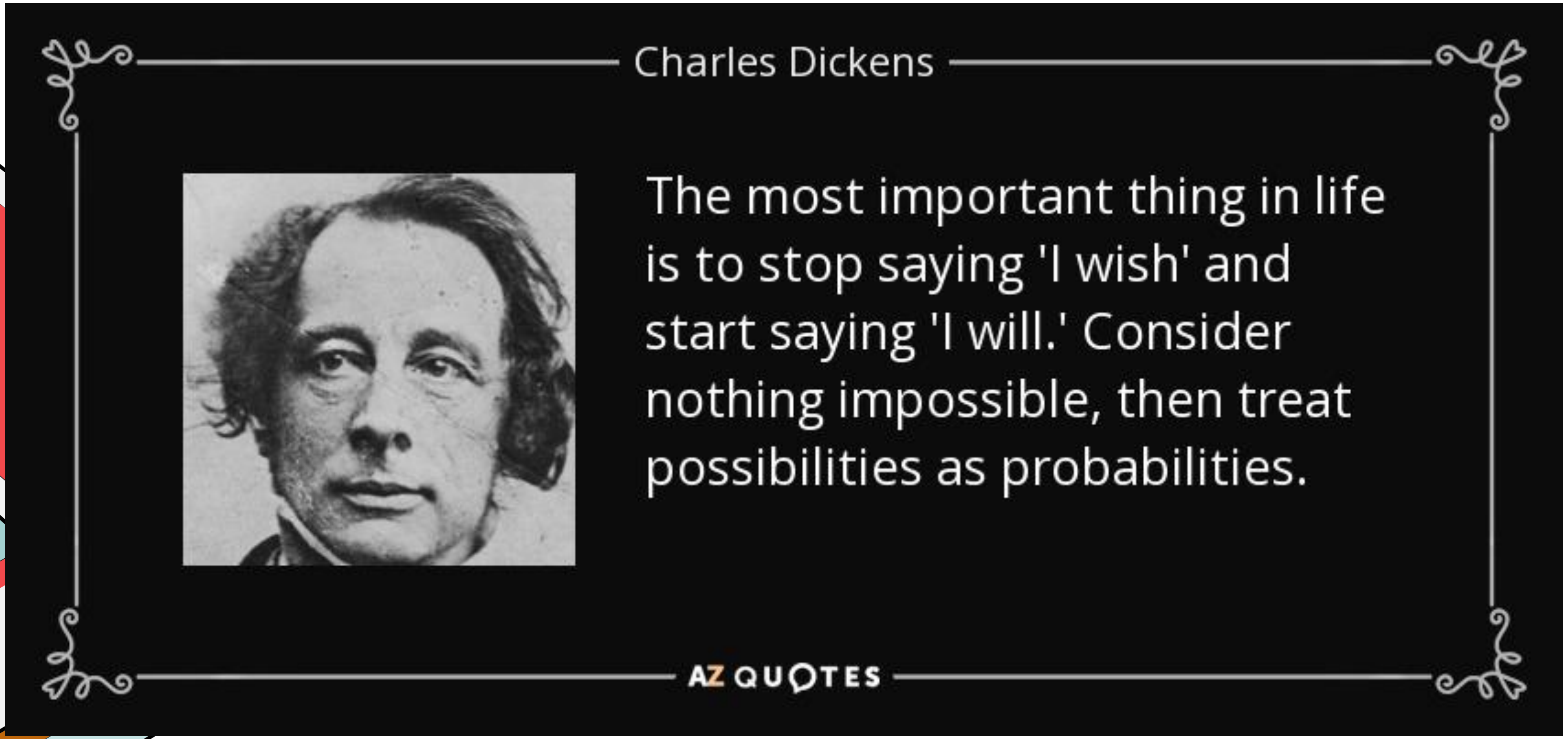
**Continuous**

- Annual income
- Height
- Temperature
- Time

# SUMMARY

| Discrete | Continuous |
| --- | --- |
| • takes on distinct values<br>• has countable (finite/ infinite) possible values<br>• counted or classified | • takes on any value within an interval<br>• has an uncountable or infinite number of possible values<br>• measured |




Charles Dickens
The most important thing in life is to stop saying 'I wish' and start saying 'I will.' Consider nothing impossible, then treat possibilities as probabilities.
AZ QUOTES

# THANK YOU